A corrected construction for unfaithful maps with two flag orbits

Bounded independent source review

12 October 2026

Abstract

We check Propositions 5.9–5.10 of Cunningham, Mochán, Montero, Potočnik and Toledo, *Faithfulness in maniplexes*, Journal of Algebraic Combinatorics 64 (2026), article 23. The printed torus construction fails the maniplex axioms. Replacing its final old color-0 operation by the old color-1 operation repairs it. The converse also needs a choice of the two flag orbits when a proposed base has repeated adjacent generators. We prove the corrected statements for possibly infinite flag sets. This reviews physical pages 23–25 and their local background, not the whole article or its final problems.

1 Conventions and the literal defect

A rank-three maniplex is a connected simple graph with one incident edge of each color $0, 1, 2$ at every flag. Its adjacency involutions are fixed-point-free, have distinct neighbors, and satisfy $r_0 r_2 = r_2 r_0$. Products below act in written order on the right. An i-face is a component after deleting color i. Flags are chain-equivalent when they have the same three faces; faithfulness means that all these classes are singletons. In class $2_{\{0,2\}}$ there are precisely two automorphism orbits; colors 0 and 2 preserve them and color 1 interchanges them. These are the source conventions.

On an old finite facet of even length $p = 2k$, let $a = r_0$, $b = r_1$ and $h = (ab)^k$. The facet is a colored cycle on $2p$ flags, with regular dihedral action $\langle a, b\rangle$ of order $2p$. Thus h is central. The printed torus formula makes the new white color 0 equal to b and its color 2 equal to ha. These commute if and only if a, b commute. If $p > 2$, distant-color commutation therefore fails. If $p = 2$, then $ha = aba = b$, so colors 0 and 2 have the same white neighbor. Consequently the printed torus construction is never a simple rank-three maniplex under its stated even-facet hypothesis.

2 The corrected construction

Let K have finite even facets, possibly of different lengths and with no finiteness assumption on its whole flag set. On $\mathcal{F}(K) \times \{0, 1\}$ define

$$\begin{aligned} (\Phi, 0)R_0 &= (\Phi a, 0), \quad (\Phi, 0)R_1 = (\Phi, 1), \quad (\Phi, 0)R_2 = (\Phi c, 0), \\ (\Phi, 1)R_0 &= (\Phi b, 1), \quad (\Phi, 1)R_1 = (\Phi, 0), \quad (\Phi, 1)R_2 = (\Phi d, 1), \end{aligned} \tag{1}$$

where $c = r_2$ and, using $h = (ab)^{p/2}$ on the old facet containing Φ,

$$d = h \quad \text{for } \widetilde{\mathrm{tr}}_P(K), \qquad d = hb \quad \text{for } \widetilde{\mathrm{tr}}_T(K). \tag{2}$$

This changes only the final r_0 in the printed torus formula to r_1. The printed projective formula is unchanged.

Theorem 1. *Both corrected constructions are connected unfaithful rank-three maniplexes. Digonal facets ($p = 2$) are allowed.*

Proof. Within an old facet, h and hb are nonidentity involutions commuting with b, and neither equals b. For a digon the four dihedral elements are $1, a, b, ab$, with $h = ab$ and $hb = a$. For longer facets a nonidentity rotation cannot be a reflection, and $hb = b$ would imply $h = 1$. The regular facet action therefore gives fixed-point-free white matchings distinct from color 0. On black flags the old color-0 and color-2 matchings already are distinct and commute. Color 1 changes layer, so coincides with neither. On white flags $[b, d] = 1$. The facet parameter is constant along a, b, so the same checks apply globally when lengths vary.

On the black layer an old a-, b-, or c-step can be followed using R_0, $R_1R_0R_1$, or R_2, respectively. Old connectivity and R_1 therefore imply new connectivity. On the white layer every word in a, b can be followed using new colors 0 and 1. In particular the distinct 2-adjacent flags $(\Phi, 1)$ and $(\Phi d, 1)$ share a facet, since $d \in \langle a, b\rangle$. As they are 2-adjacent, they already share their vertex and edge. They are chain-equivalent, proving unfaithfulness. □

3 Regularity and the exact exceptions

Suppose K is regular. Its flags can be identified with a group $H = \langle a, b, c\rangle$, where the three marks are distinct nonidentity involutions, $[a, c] = 1$, and $p = |ab| = 2k < \infty$. The new flags are $H \times \{0, 1\}$; (1) uses right multipliers, and left multiplication by H acts transitively on each layer.

Theorem 2. *The corrected projective construction is regular exactly when $c = h$; the corrected torus construction is regular exactly when $c = ha$. Otherwise its full automorphism group is the displayed H and its type is $2_{\{0,2\}}$. The exceptions are precisely the source's marked $P\{k\}$ and $T\{k\}$ families where these are simple maniplexes. Thus $P\{1\}$ is allowed, but $T\{1\}$ would have $c = b$ and is excluded by simplicity.*

Proof. An automorphism of a connected colored graph is determined by one flag image, by following finite paths. This holds for infinite graphs too. Hence every layer-preserving automorphism equals a unique displayed left multiplication. If any automorphism interchanges the layers, compose with a left multiplication to obtain τ sending $(1, 0)$ to $(1, 1)$. Since it commutes with R_1, it also sends $(1, 1)$ to $(1, 0)$; together with H it is flag-transitive.

The word $W = (R_0R_1)^p$ multiplies by h on both layers. Indeed, $(R_0R_1)^2$ multiplies black coordinates by ab and white ones by ba; their kth powers coincide. In the projective case R_2 and W agree at $(1, 1)$. Commutation with τ transfers that agreement to $(1, 0)$, so $c = h$. For the torus case use agreement of R_2 and WR_0 at $(1, 1)$; its transfer gives $c = ha$.

Conversely either equality gives $H = D = \langle a, b\rangle$. The dihedral automorphism θ interchanging a, b fixes h. In the projective case it interchanges $c = h$ and $d = h$; in the torus case it interchanges $c = ha$ and $d = hb$. Thus $(g, i) \mapsto (\theta(g), 1 - i)$ commutes with (1), proving regularity. The source's defining relations on physical page 19 are exactly these descriptions of $P\{k\}$ and $T\{k\}$. □

The proof of source Proposition 5.9 labels the displayed quotient $2_{\{1,2\}}$. Its actual colors give $2_{\{0,2\}}$, as in the statement: only color 1 changes layer. This is a notation slip, not a further family.

4 The converse and a degenerate proposed base

Lemma 3. *Let a, b be distinct involutions in any group. If $d \in \langle a, b\rangle$ commutes with b and $d \notin \{1, b\}$, then $p = |ab|$ is finite and even, and $d \in \{h, hb\}$ where $h = (ab)^{p/2}$.*

Proof. Put $t = ab$. Dihedral normal forms are t^j and t^jb, with $btb = t^{-1}$. They do not overlap: if $b \in \langle t\rangle$, the group would be cyclic and generated by two involutions, forcing $a = b$. Either form commutes with b exactly when $t^{2j} = 1$. If t has infinite order or odd finite order this gives only $1, b$. With finite even order the two further possibilities are h, hb. □

Theorem 4. *Every unfaithful rank-three maniplex of full type $2_{\{0,2\}}$ is, up to duality, one of (1)–(2) from a regular simple rank-three maniplex with finite even facets. One may need to exchange the names of the two flag orbits before choosing the base. There is no additional exceptional family.*

Proof. We first reprove the rank-three observation used in source Proposition 3.4, which attributes it to [12, Lemma 3.1]. A chain-equivalent flag lies in the same edge, whose flags are $\Phi, \Phi^0, \Phi^2, \Phi^{0,2}$. If only the last is initially known to be equivalent to Φ, it lies in the same facet, and its 0-neighbor Φ^2 lies there too. Thus $\Phi^2 \approx \Phi$. In every nontrivial case, up to duality, there is a 2-adjacent equivalent pair.

Name its orbit white, choose a root w there and put $v = w^1$ in the black orbit. Set $H = \mathrm{Aut}(M)$. Each orbit is a free H-orbit, since a flag stabilizer is trivial. Identify gv with $(g, 0)$ and gw with $(g, 1)$. There are unique nonidentity involutions $a, c, b, d \in H$ giving the appropriate 0- and 2-neighbors of v, w. They are involutions because their squares fix the base flag. All adjacencies now have (1), and simplicity and distant commutation give

$$a \neq c, \qquad b \neq d, \qquad [a, c] = [b, d] = 1. \tag{3}$$

Connectivity implies $H = \langle a, b, c, d \rangle$: reading a path multiplies coordinates by these elements, and each is realized by a path ending in its starting layer (using R_1 before and after a white operation if necessary).

The facet through v, w is $D \times \{0, 1\}$, where $D = \langle a, b \rangle$. Since $w^2 \approx w$, we have $d \in D$. If $a = b$, then $D = \{1, b\}$, contradicting (3). Lemma 3 therefore gives finite even $p = |ab|$ and $d = h$ or hb. Consequently $H = \langle a, b, c \rangle$.

If $b \neq c$, the marks a, b, c are pairwise distinct nonidentity involutions with $[a, c] = 1$. Their colored Cayley graph is a regular simple maniplex K, and (1) is its corresponding corrected construction. This uses neither a finite ambient group nor the intersection property of a string C-group.

The only possible duplicate is $b = c$, omitted in the source's base sentence. Then $[a, c] = 1$ gives $[a, b] = 1$, so $p = 2$, $H = D \cong C_2^2$ and $h = ab$. We have $d = ab$ or $d = a$. If $d = a$, interchanging a, b in C_2^2 and interchanging the layers gives a color-preserving automorphism. That would make M regular, contrary to its hypothesis. Hence $d = ab$. Exchange the layer names. The four marks become

$$(a', b', c', d') = (b, a, ab, b).$$

Now a', b', c' are the three distinct nonidentity elements of C_2^2, and $h' = a'b' = ab$ satisfies $d' = h'b'$. The base is the colored complete graph K_4 and the construction is the corrected torus version. The new white pair still shares a facet because $d' \in \langle a', b' \rangle$. This closes the duplicate-generator case. □

5 Eight flags refute the printed classification

Take $H = C_2^2 = \{1, x, y, xy\}$ and (1) with

$$(a, b, c, d) = (x, y, y, xy). \tag{4}$$

The layer matchings are distinct and commuting; color 1 changes layer, and x, y generate all coordinates. Thus this is a simple connected maniplex. The facet group $\langle x, y \rangle$ and vertex group $\langle y, xy \rangle$ both equal H. Each layer is one four-flag edge. There is therefore one vertex, two edges and one facet, with the layers as chain-equivalence classes.

The word relation $R_1R_0R_1 = R_2$ holds at every black flag because $b = c = y$, and at no white flag because $a = x \neq d = xy$. No color-preserving automorphism can interchange layers. Left H supplies all four layer-preserving automorphisms, so the full symmetry type is genuinely $2_{\{0,2\}}$.

If this eight-flag maniplex, or its dual, were a printed construction, its base would have four flags. Every simple properly 3-colored cubic graph on four flags is K_4, with its three perfect matchings as colors. The printed torus construction is invalid by Section 1; the projective construction from K_4 is regular by Theorem 2, since its third mark is $ab = h$. Neither equals (4) or its dual. Thus literal

Proposition 5.10 fails. After the formula repair and layer choice of Theorem 4, (4) is the corrected torus construction from K_4, not a new family.

The accompanying finite checks are optional diagnostics, not premises for the infinite statements. No Lean formalization is asserted. The following source paragraph on exact chain-class sizes, and other sections of F007, remain outside this bounded review.